A Reusable Georeferenced Spatial Benchmark Layer for the IEEE 30-Bus Test System

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Abstract—This paper describes a project-authored, synthetic georeferencing of the IEEE 30-bus test-system topology as a reusable spatial benchmark layer. The release contains 30 bus points, 41 branch routes, stable public identifiers, directed endpoint crosswalk fields, a GeoJSON schema, a manifest, deterministic release packaging, and machine-executable quality controls. It is intended to make a familiar electrical test topology inspectable by spatial and geospatial methods without implying that the coordinates or routes describe an operating utility system.

The current pre-release is supported by spatial-integrity and release-integrity evidence, not by wildfire, AC optimal power flow, reliability, or operational-performance validation. The executed validator reports a connected 30-node graph, 34 line routes and 7 link routes, zero invalid/non-simple/zero-length geometries, exact route-endpoint coincidence under a 0.01-m criterion, and 687.388234 km total route length after transformation to EPSG:3310. Wildfire–grid modeling is presented only as one possible downstream use. The manuscript records the dataset boundaries, reproducible methods, verification evidence, and remaining requirements for a citable archived release.

Index Terms—IEEE 30-bus system, georeferenced power systems, GeoJSON, reproducible research, spatial benchmark, synthetic infrastructure

I. Introduction

Spatial methods increasingly require electrical-network examples with explicit geometry. Yet an electrical test case alone does not specify a reusable point-and-route spatial representation, a coordinate reference contract, stable spatial identifiers, or quality checks for reconciling routes to electrical endpoints. The IEEE 30-bus topology is widely used for power-system studies; MATPOWER distributes a standard machine-readable implementation of it [1]. A recent wildfire progression and grid-risk study used an IEEE 30-bus case mapped into a California study area [2]. That work motivates spatial interfaces, but its wildfire methodology and results are not the contribution evaluated here.

This manuscript documents a separate project-authored spatial asset: a synthetic, inspectable georeferencing of the 30-bus topology. The contribution is deliberately bounded. It is not a claim that the benchmark reproduces real transmission corridors, validates fire behavior, estimates risk, or improves grid operation. Rather, it supplies a stable spatial layer that method developers can use, inspect, replace, or extend while retaining an explicit topology crosswalk and a reproducible validation trail.

The dataset follows GeoJSON conventions [3] and stores coordinates in WGS84 longitude/latitude. All reported metric geometry checks are conducted after transformation to California Albers (EPSG:3310), avoiding the use of angular degrees as lengths. The release is hosted with code, documentation, checksums, and tests in a public repository [4]. A DOI has not yet been minted; therefore this manuscript is a working IEEEtran draft rather than a submission-ready data descriptor.

II. Dataset Scope and Design

A. Asset boundary

The release is a synthetic spatial benchmark. It contains bus locations and branch-route geometries for the selected IEEE 30-bus topology. The coordinate placement and corridors are project-authored and illustrative. They must not be interpreted as locations, ownership, line characteristics, or operational states of a real electric utility.

The dataset excludes third-party payloads unless separately cleared. In particular, it does not redistribute IEEE case files, FARSITE executables or inputs, environmental landscape/weather inputs, or IEEE Access article text, figures, or tables. The previous study is cited for context [2]; it is not copied into this release. Such separation is essential for both provenance and reusable dataset scope.

B. Data model

The principal asset is data/network/canonical-network.geojson, a GeoJSON FeatureCollection. Point features represent buses; LineString features represent branch routes. Each bus has a numeric bus field and an immutable zero-padded bus˙id such as bus˙01. Each route has numeric branch, immutable branch˙id such as branch˙01, directed fbus/tbus endpoint fields, and a normalized branch˙type of line or link. The type field preserves the distinction between 34 line routes and 7 transformer-type links.

The schema enforces geometry-specific fields and identifier forms, while the manifest records version, coordinate contract, generation provenance, and cryptographic file hashes. A data dictionary and datasheet accompany the release. Figure 1 is a view of the released synthetic layer, not a claim of real-infrastructure mapping.

Working manuscript, version 0.1.0-pre-release. Author list, affiliations, funding, and acknowledgments require confirmation before submission. The release described here is a project-authored synthetic spatial layer; it does not represent real utility infrastructure.

TABLE I
Released spatial-layer composition and identifier contract.

Entity	Count	Stable public identifier
Bus point	30	bus˙01–bus˙30
Branch route	41	branch˙01–branch˙41
Line route	34	branch˙type=line
Link route	7	branch˙type=link

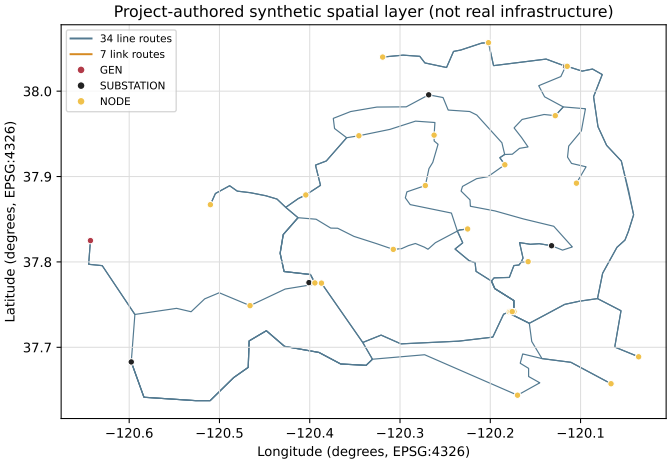


Fig. 1. Project-authored synthetic bus points and branch routes in stored WGS84 coordinates. The figure is a dataset depiction, not a map of utility infrastructure.

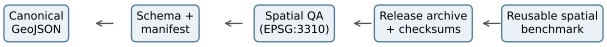


Fig. 2. Release workflow. Boxes represent artifacts and executed checks; the diagram does not represent a performance-evaluation pipeline.

III. Reproducible Construction and Validation

A. Construction workflow

A project script constructs the canonical layer from a frozen, documented input reference, reconciles route orientation to the declared fbus→tbus direction, and adds the public identifier and branch-type fields. A documented correction resolves a branch-24 source-label inconsistency. The workflow snaps only source route endpoints within a declared 500-m construction threshold to their associated bus coordinates; this is a construction rule, not a measure of physical line accuracy.

The release archive is manifest-first. It includes the canonical GeoJSON, schema, provenance/readme material, validation code, selected tests, release manifest, and SHA-256 checksums. The deterministic builder updates member hashes before constructing the archive. Figure 2 summarizes this traceable release path.

TABLE II
Executed pre-release verification evidence.

Check	Observed result
Schema validation	0 errors
Graph connectivity	One component containing all 30 buses
Route geometry	0 invalid, non-simple, or zero-length routes
Endpoint alignment	Maximum 0.0 m; criterion $\leq$0.01 m
Metric route length	687.388234 km total; EPSG:3310
Automated test suite	25 tests passed at the recorded pre-release build
Release-package validator	Passed; archive checksum recorded in repository

B. Technical validation protocol

The validation procedure operates on the released GeoJSON rather than on screenshots or manually copied values. It checks: (i) the 30/41 feature counts and stable-ID domains; (ii) allowed branch-type domains; (iii) JSON Schema conformance; (iv) valid, simple, nonzero-length LineStrings; (v) transformed endpoint distance from each route endpoint to the referenced bus point; (vi) graph connectivity induced by fbus/tbus; and (vii) metric route lengths after EPSG:4326-to-EPSG:3310 transformation with explicit longitude/latitude axis ordering.

Let p_i be the projected coordinate of bus i and $e_{ij}^{(s)}$, $e_{ij}^{(t)}$ be the projected start and end coordinates of route (i, j). Endpoint reconciliation evaluates

$$d_{ij} = \max\{\|e_{ij}^{(s)} - p_i\|_2, \|e_{ij}^{(t)} - p_j\|_2\}, \tag{1}$$

with acceptance criterion $\max d_{ij} \leq 0.01$ m. Route length is computed as the planar length of each EPSG:3310-transformed LineString; it is not inferred from the source length or Shape˙Length attributes, whose units are undocumented.

IV. Executed Verification Evidence

Table II reports the current executed pre-release evidence. These checks establish internal spatial and release integrity for this particular synthetic payload. They do not establish external geodetic accuracy, real-corridor fidelity, wildfire model validity, ACOPF validity, risk-prediction performance, or operational suitability.

The 687.388234-km total differs from the 702-km figure described for the prior case study [2]. The release treats this as a disclosed spatial-layer difference, not as an error to be forced away and not as evidence of superiority. Any downstream analysis must report which geometry version and metric procedure it uses.

V. Use, Limitations, and Release Requirements

Appropriate uses include topology-aware GIS demonstrations, spatial-overlay methods, reproducible teaching examples, and interfaces for independently licensed wildfire, weather, fuel, or grid-analysis components.

Users may replace compatible inputs or add method-specific attributes, provided they retain provenance, avoid reusing immutable IDs for altered features, and document changes.

The benchmark has important limits. It is synthetic. It contains neither terrain/fuel/weather data nor fire simulations; it does not validate wildfire spread, component exposure, dispatch, outages, power flow, or risk. Straight or source-derived route geometry should not be interpreted as surveyed corridors. The present GitHub release is public but pre-DOI. Before an archival release and manuscript submission, the project requires an independent cold-start reproduction, an external technical review, final author/funder/affiliation metadata, a release tag, an archived DOI, and a final provenance/rights review.

VI. Conclusion

This work packages a narrow but useful research asset: a transparent, synthetic georeferencing of the IEEE 30-bus topology with stable identifiers and executable spatial quality controls. Its evidence supports claims of internal spatial consistency and reproducible release packaging only. By keeping wildfire-grid modeling as an optional downstream use, the dataset can support a broader class of spatial-method studies while avoiding unsupported operational or physical claims. The next release stage is independent reproduction and archival publication, followed by revision of this draft against documented reviewer feedback.

Data and Code Availability

The pre-release dataset, code, validation report, archive manifest, and tests are available at https://github.com/BehrouzSohrabi/georeferenced-ieee30-wildfire-benchmark. The release is not yet assigned a DOI. Third-party inputs are excluded or pointer-only as documented in the repository.

Acknowledgment

To be completed after confirmation of contributors, institutional affiliations, funding, and independent reviewers.

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